Resume Cover

EDUCATION

University of Pennsylvania (UPenn), Biology and Computer Science, Chemistry minor **GPA:** 4.0/4.0
Illinois Math and Science Academy, Chicago, IL **GPA:** 3.94/4.0

AWARD HIGHLIGHTS

Grant Winner: **Wharton Venture Innovation Fund** | *IVision: UNet Deep Learning for vein visualization in clinical settings*

1st Place Winner: **National Junior Science and Humanities Symposium** (JSHS) in the Math and Computer Science Category (8,000+ initial competitors) | *Natural language processing, signal processing, and machine learning to understand how DBS can be used in clinical settings using 38 epilepsy patients*

Top Score: **SIMIODE International Challenge Using Differential Equations Modeling** in Problem C (~500 undergraduate and high school competitors) | *created a mathematical model based on calculus-based physics equations to answer mathematical modeling questions on dog behavior*

RESEARCH HIGHLIGHTS

1st Author Peer-Reviewed Research Publication: "Differentiating hand gestures from forearm muscle activity using machine learning." **International Journal of Occupational Safety and Ergonomics** (Q1 Journal), Taylor & Francis, August 2024. doi.org/10.1080/10803548.2024.2383021**.**

1st Author Peer-Reviewed Research Publication: "Investigating brain activity patterns during learning tasks through EEG and machine learning analysis." **International Journal of Information Technology, Springer**, April 2024. doi.org/10.1007/s41870-024-01856-4**.**

Patent Co-Inventor: "Visual assistance system." Microcontroller and sensor development to assist the deaf/blind community. September 2026. Patent information.

1st Author Conference Proceedings and Oral Presentation: "Neural Understanding of STEM Activities." **IEEE International Symposium on Computer Science and Educational Technology**, Laubusch, Germany, July, 2024. doi: 10.1109/ISCSET58624.2024.10807951.

EMPLOYMENT

Research Assistant at Penn Center for Neuroengineering and Therapeutics (Dec 2025 - Present)

- UPenn CURF 2026 Summer Grant Recipient
- Analyzed a large cohort of 1,300 epilepsy patients using signal processing, machine learning, and statistics, to build bedside tool for neurologists (github and github)
- Created a GUI for clinical annotation of an interictal spike localization method on 200 epilepsy patients (github)
- Analyzing lateralized periodic discharges along the ictal-interictal continuum (IIC) and seizure onset zone machine learning algorithms

Website Lead at the International Youth Neuroscience Association (October 2023 - May 2025)

- Solely rebuilt the entire website from WIX to being a programmable website that lowered costs by ~$300 annually (https://youthneuro.org/) and creating Lottie animations and design templates for the site